Inner Products and the Exponential Function

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1 Inner Products, Norms and the Cauchy–Schwarz Inequality

Throughout this section, let V be a vector space over the field $\mathbb{R}$. An *inner product* on V is a map $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{R}$ that is linear in each argument, symmetric, and positive definite, meaning that $\langle x, x \rangle \geq 0$ for all $x \in V$, with equality if and only if $x = 0$.

Every inner product induces a *norm* and, through it, a *metric*:

$$\|x\| \coloneqq \sqrt{\langle x, x \rangle}, \qquad d(x, y) \coloneqq \|x - y\|.$$

For a nonzero vector $x \neq 0$, we call $\hat{x} \coloneqq x / \|x\|$ the associated *unit vector*, which satisfies $\|\hat{x}\| = 1$.

The estimate relating the inner product to the norm is the *Cauchy–Schwarz inequality*

$$|\langle x, y \rangle| \leq \|x\| \, \|y\|,$$

which is satisfied with equality if and only if x and y are linearly dependent. It guarantees that the angle θ between two nonzero vectors is well defined through

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \, \|y\|},$$

and it implies the *triangle inequality* $\|x + y\| \leq \|x\| + \|y\|$. A norm that comes from an inner product is special: it obeys the *parallelogram law*

$$\|x + y\|^2 + \|x - y\|^2 = 2 \, \|x\|^2 + 2 \, \|y\|^2.$$

Given an orthonormal system (i.e., a system of vectors that are pairwise normal and of unit length) $e_1, \ldots, e_n$, the *orthogonal projection* of x onto its span is

$$\operatorname{proj}_U(x) = \sum_{i=1}^{n} \langle x, e_i \rangle \, e_i,$$

where $(x - \operatorname{proj}_U(x))$ is orthogonal to every e_i. Comparing the two contributions yields *Bessel's inequality*

$$\sum_{i=1}^{n} |\langle x, e_i \rangle|^2 \leq \|x\|^2.$$

2 The Exponential Function as a Power Series

The real *exponential function* can be defined by the power series

$$\exp(x) \coloneqq \sum_{n=0}^{\infty} \frac{x^n}{n!}, \tag{1}$$

which converges for every $x \in \mathbb{R}$. It satisfies $\exp(0) = 1$ and the functional equation $\exp(x + y) = \exp(x)\exp(y)$, so in particular $\exp(x)\exp(-x) = 1$.

There is also a connection to a classical sequence limit

$$\exp(x) = \lim_{n\to\infty}\left(1 + \frac{x}{n}\right)^n, \qquad \text{giving} \qquad e \coloneqq \exp(1) = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n.$$

Additionally, the exponential function is its own derivative,

$$\frac{\mathrm{d}}{\mathrm{d}x}\exp(x) = \exp(x),$$

which is very sensible given the original form in equation (1), and, together with the initial value $\exp(0) = 1$, characterises it uniquely among all differentiable functions.